

The Peril and Potential of XR-based Interactions with Wildlife

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ABSTRACT

In “Being a Beast”, Charles Foster recounts living with, and as, wildlife (e.g., otters, foxes). These encounters, he contends, forge human-nature connections which have waned, negatively impacting biodiversity conservation. Yet, we need not live amidst beasts to bridge the human-nature gap. Cross-reality (XR) platforms (i.e., virtual and augmented reality) have the unique capacity to facilitate pseudo interactions with, and as, wildlife, connecting audiences to the plight of endangered species. However, XR-based wildlife interaction, I argue, is a double-edged sword whose implementation warrants as much attention in HCI as in environmental science. In this paper I highlight the promise of XR-based wildlife encounters, and discuss dilemmas facing developers tasked with fabricating mediated interactions with wildlife. I critique this approach by outlining how such experiences may negatively affect humans and the survivability of the very species seeking to benefit from them.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction**; • **Interaction paradigms**; • **Virtual reality**; • **Mixed / augmented reality**;

KEYWORDS

Virtual reality, augmented reality, biodiversity, conservation, nature

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1 THE HUMAN-NATURE GAP

“In the so-called information age, humans are even more out of tune [with nature]. As during the First Industrial Revolution, technology is a primary suspect” [40].

Humans have a hardwired inclination to form bonds with other lifeforms, human or otherwise, a phenomenon described by the biophilia hypothesis [45]. Despite humanity’s evident ties to the natural world, a growing fissure between (wo)man and nature has been observed over the past century. Nature Deficit Disorder (NDD) was a term coined to describe this schism, one attributed to rapid urbanization, increased media consumption, and reduced time spent in natural environments engaging with wildlife. It is ironic, then,

that our species has never spent less intentional time interacting with the natural world, and yet our collective influence over the well-being of the planet has never been greater nor more detrimental [71].

Intergovernmental panels recognize the gravity of NDD and its implications for the preservation of the natural world, leading them to emphasize that preserving biodiversity is a priority and responsibility shared by nation states, its researchers, and non-academics [42]. Reducing the human-nature gap is proposed as an integral step towards preserving biodiversity, an outcome predicated on encouraging connectedness with nature (CWN), a subjective sense of closeness with the environment known to contribute to pro-environmental outcomes, such as awareness and protection [17, 63], and even well-being [55]. However, it is less a question of whether aspirations for CWN are merited, and more a question of how media technology, the proposed culprit of NDD, may play a role in spurring connectedness. As human-computer interaction (HCI) scholars and content creators, shepherds of virtual landscapes, it is imperative to first acknowledge that there is collateral damage associated with increasing the share of human life spent engaging with synthetic rather than organic nature/wildlife through media. Second, and perhaps more importantly, I believe that it is our field’s collective duty to leverage emerging technology to tailor such engagements so as to create net gains for biodiversity conservation measures.

To this end, this paper serves as a critique of cross-reality (XR) platforms (i.e., virtual and augmented reality) and their use as conduits for human-wildlife interactions. I review the promise of XR for biodiversity conservation, highlight existing XR experiences that place users face-to-face (FtF) with various species, and discuss the myriad of ethical and ecological downsides that may emerge from their widespread adoption. It is my hope that this paper can spark discussion between and within HCI, media psychology, and environmental science communities on how to responsibly curate XR-based wildlife experiences.

2 TECHNOLOGICAL NATURE

The axiological foundations of HCI and media studies have long been questioned as they relate to mediated experiences with nature, or as Kahn describes it, “technological nature” [28, 33]. Studies on this topic have examined how nature is depicted and treated on television [75] and film [54], and how such depictions may ultimately influence conservation outcomes considering audiences may experience nature partly through such channels. Similarly, video games that simulate hands-on nature experiences, such as animal husbandry and farming, have brought into question how such pseudo interactions with nature may indirectly shape CWN and other environmental outcomes. For example, studies have examined how farming games, such as Lil’ Green Patch [27], and interactions with virtual nature in games like World of Warcraft [73] shape CWN. Specific to mediated interactions with wildlife,

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scholars have also examined the impact of playing with virtual [47] and robotic animals [43] on human-animal bonds. Where environmentalists once expressed concerns of a time “when screens replace backyards” [20], today we see screens replacing farms, zoos, and entire worlds.

Despite the prevalence of technological nature and ambiguity surrounding its long-term influence over CWN, I believe environmentalists have been able to quell concerns pertaining to the synthetic supplanting the organic. That is, the data generally suggests that, as it relates to human-wildlife interactions, “there’s nothing like the real thing” [30]. At the crux of this axiom are two important assumptions: First, that human-wildlife connectedness is forged through direct interactions comprised of multisensory information (e.g., audiovisual, haptic, smell), and second, that an organic sensory experience cannot be equated to synthetic proxies, nor wholly replicated through contemporary media platforms [33, 72]. Put simply, it is difficult to replicate the sensory richness associated with coming FtF with a wild animal. Indeed, research shows that direct nature-related experiences (e.g., birdwatching, taking photos of nature) contribute to conservation outcomes to a greater extent than indirect, or mediated, engagement (e.g., watching nature programs on TV) [58]. However, advances in visual computing and sensor technologies have dovetailed to increase the capacity with which mediated experiences, with wildlife or otherwise, can look and feel “like the real thing”. In blurring the line of demarcation between synthetic and organic human-wildlife interactions, these advances bring into question how mediated interactions with wildlife may come to shape the future of our planet and our connection with it.

3 HUMAN-WILDLIFE INTERACTIONS IN XR

It is human nature to actively seek to replace a “real thing” with an alternative [30], and media-based substitutes can effectively satiate certain experiential needs. Indeed, the media equation paradigm has long argued that humans are limited in their ability to distinguish between mediated and real representations [57]. No more is this evident than in the emergence of XR platforms, such as virtual reality (VR) and augmented reality (AR), where mediated interactions with virtual spaces and beings are capable of eliciting psychological and physiological responses akin to their in vivo counterparts [4, 29].

3.1 Virtual Reality

The perceptual equivalence of VR is rooted in the platform’s ability to elicit a subjective sense of “being there” (spatial presence) [69] and “being with” (social presence) [48] beings in a computer-generated virtual environment. Using head-mounted displays (HMDs) and additional peripherals (e.g., hand controllers, haptic feedback devices), users’ senses are immersed into alternate realities where they can assume the perspective of another virtual body (avatar). In embodying and controlling this new self-representation, users can subsequently feel the subjective sense that their virtual body is their own. This illusion, referred to as body transfer (BT) [5] or body ownership illusion (BOI) [8, 66], contributes to connectedness, or a self-other overlap, with the avatar and the group it represents (e.g., gender, race). More importantly, this perceptual illusion is achievable with humanoid bodies [41], animal bodies [2, 34–36], and even trees [22]. These affordances

ultimately contribute towards a sense of non-mediation, and the consistent finding that humans perceive and respond to virtual beings, including other people [32] and animals [70], in ways comparable to in vivo exposure.

3.2 Augmented Reality

Unlike VR, AR experiences embed virtual content onto the user’s real-world environment, either using an HMD, smartphone, or other mobile device. That is, where a VR experience can transport a user to a virtual jungle alongside a virtual panther, a similar AR experience would place the panther in the user’s living room. Despite their technical differences, a similar degree of realism and presence can be achieved through AR modalities. Users can similarly feel a sense of social presence with virtual characters placed in their space, and respond to them as they would in real life (IRL) [44]. Specific to animals, previous work has found that human responses to insects [13] and arachnids [70] in AR are comparable to in vivo exposure. Indeed, it is this equivalence that has led scholars to note that games like “Pokémon Go!”, an AR-based game about capturing fictional wildlife, may contribute to biodiversity efforts by implicitly connecting humans to non-human life [19].

3.3 Purpose of this Work

The case for XR-based wildlife interactions as a method to connect humans to nature, bridge the human-nature gap, and subsequently contribute to biodiversity conservation is rather straightforward:

- Individuals exhibit a greater tendency to protect species with whom they feel connected with.
- Connectedness is strongest when forged by direct rather than indirect interactions with wildlife.
- Accessibility to direct interactions with wildlife, especially threatened species, are extremely limited.
- XR-based interactions with wildlife can provide the perceptual illusion of direct experience.

As a cautionary note, Melson argues, “one cannot rule out the possibility that increasing exposure to mediated interactions with animals, through robotics, virtual reality and other media, may come at the expense of direct engagement with living animals” [43]. Research on the use of XR to connect humans to the plight of wildlife is still in its infancy. As such, whether the juice is worth the squeeze remains to be seen. To reconcile this ambiguity, and spur discourse on how XR can be responsibly employed to benefit biodiversity, this paper highlights several projects using XR-based human-wildlife interactions for conservation efforts. Specifically, I delineate across two primary forms of XR-based interactions with wildlife. The first is “interactions with wildlife”, which details XR-based experiences that allow users to engage with animals as humans. The second is “interactions as wildlife”, which details XR-based experiences that allow users to embody animals. After reviewing each approach and associated examples, I discuss potential ethical, social, and ecological issues associated with their widespread use.

4 INTERACTIONS WITH WILDLIFE

Ironically, the institutions most concerned with how technology is widening the human-nature divide, such as zoos and aquariums,

are increasingly using XR technology to facilitate mediated interactions with wildlife. Perhaps the most popular and cost-effective use involves distributing 360-degree video of their enclosures. For example, the North Carolina Zoo created “Expedition Africa VR” to give audiences an up close and personal look at species like Rhinos and Elephants. The value proposition is simple: when wildlife is out of reach or out of sight, any access is better than no access, regardless of how diluted that access is comparable to the real thing. Zoos cannot guarantee optimal views of their animals, nor can they allow visitors to get within close proximity to certain species. An immersive experience like Expedition Africa VR allows Zoos to place visitors, and those without the financial or physical ability to enter their premises, face-to-face with wildlife.

Environmental organizations proclaim that, “instead of using technology to escape, now we can connect to the real world inside the mobile landscape, allowing visitors to more actively engage with animals and exhibits in a familiar, fun and exciting way” [77]. However, the vast majority of these experiences leverage 360 video as opposes to VR or AR platforms. As a result, audiences cannot directly interact with representations of wildlife, just merely observe them. There are a few exceptions, especially as they relate to extinct species. For example, the Florida Museum of Natural History debuted an AR installation allowing users to see themselves alongside Sabretooth Tigers and Mammoths. However, the creatures are not spatially aware of visitors in the environment, and thus even simple interactions between man and beast (e.g., mutual gaze) are not possible. Moreover, one would be hard pressed to find a similar experience for non-extinct, but altogether threatened, wildlife.

4.1 Why Don't We Have Noah's Ark in XR?

There are several reasons why zoos and aquariums may not be investing in replicating virtual wildlife. For one, the cost of creating interactive human-wildlife experiences is high. One of the major costs involved is the creation of accurate 3D models of wildlife. While scholars note that advances in photogrammetry and open-source software are streamlining the process of creating 3D models for use in XR experiences, animating these creatures to ensure accurate and realistic motion remains a barrier [12]. Indeed, the craft of 3D animal tracking is relatively new [10], and even collecting motion data from actual animals is considered extremely difficult [62]. Moreover, because these processes require access to live animals with which to capture spatial and motion data, and threatened wildlife are difficult to gain direct access to, accurate 3D modeling is limited to species that are accessible and non-endangered, undermining the value of XR-based interactions with threatened wildlife.

Moving forward, costs associated with (re)creating virtual wildlife for use in XR-based interactions will surely go down. For example, emerging computer vision frameworks are capable of automatically extracting 3D textured models from images of wildlife [78]. Additionally, advances in video-based marker less motion capture technology provide inexpensive methods for capturing and integrating animal movement onto virtual wildlife without necessitating access to live animals [37]. Given the inevitable trajectory towards more human-wildlife interactions via XR, the following sections highlight several projects at various stages of development

that seek to explore the implications of such experiences on CWN and biodiversity conservation.

4.2 Cleaning Penguins in VR

For much of the 20th century, an average of 357 African Penguins alone were caught in oil spills [18] every year. Marine scientists note that conservation of penguin species, as well as most marine life, requires the communication of successful conservation efforts [60]. With noble intentions, I sought to leverage XR as a means by which to communicate how these threats affect penguin species, and how humans can actively take actions to mitigate them. To accomplish this, a VR simulation was created for the Oculus Rift HMD which transported users to Antarctica where they were onboarded as volunteers to help rehabilitate oil-slicked penguins. Through the interactive experience, users were brought FtF with threatened wildlife, and were able to clean the oil off several penguins by using their hand controllers (Figure 1).

Variations of the simulation have been tested both in controlled lab settings and in the field (i.e., science centers). Across these sites one thing was found to be certain: these experiences can have profound effects, at least in the short term, on how humans perceive and advocate for threatened wildlife. Attitudes towards the species and advocacy intentions benefit from exposure to wildlife in VR, at least when interactions involve users helping wildlife, as opposed to unstructured interactions. Moreover, a moderate to high level of social presence can be elicited through these experiences, which is particularly promising considering that social presence is a multidimensional construct which can be characterized, at least in part, by a cognitive and affective connection with a target [48]. I will not harp on the potential benefits of this experience, the upside of XR has been established. Rather, it is important to elaborate upon how any conservation-related gains accrued through such interactions in VR may work counter to the underlying goal of advancing biodiversity efforts.

4.2.1 Soap Washes Away Oil, Not Biases. The penguin simulation, and by extension many VR-based content seeking to connect humans to the plight of wildlife, can be a double-edged sword in the fight for biodiversity conservation. Humans extend differential concern and attraction towards species based on various factors, including attractiveness, ornamentation, and size [46]. Regarding the latter, the penguin simulator demonstrated that these biases extend to how we treat virtual wildlife of different sizes as well, even within the same species [53]. Specifically, users demonstrated a caregiving preference towards larger rather than smaller penguins, influencing how long users spent rehabilitating penguins as well as the total number of penguins they helped. This ultimately raises legitimate questions about wildlife representation in XR-based interactions with threatened species, especially if users are provided with a conservation action upon completion (e.g., donations). What size animal do we use when creating these experiences? Where do the ceiling effects reside among species with large versus small size variability, such as bees and elephants, respectively?

4.2.2 This is not “Happy Feet”: Anthropomorphizing Wildlife. In a separate field study using an adapted version of this experience for the Oculus Go, users cleaned penguins using a water gun to



Figure 1: A VR user cleaning a virtual oil-slicked penguin using the Oculus Rift (left) and an AR user rehabilitating a penguin using the Magic Leap HMD (right).

compensate for the absence of 6 degrees of freedom afforded by the HMD. The study compared two versions of the simulation containing penguins of the same size, though one either displayed a dialogue box thanking the user for their help and the other was absent this cue. The anthropomorphizing of any animal involves this kind integration of human-like behavior, and has long been debated since scholars questioned whether the Tamagotchi would undermine real interpersonal relationships among children [11].

As O'Rourke asserts, "the 'real' is not only affected by the virtual, but is actually constructed by it" [47]. Thus, the anthropomorphizing of these creatures may similarly undermine the purpose of the experience, which is to connect users to the plight of the species in ways that spillover into real-world actions.

4.2.3 Context Matters: Balancing Enjoyment with Impact. Both penguin simulations were rated moderately high in terms of enjoyment by users. However, it is evident that in vivo exposure to suffering wildlife is anything but enjoyable. While volunteers may derive fulfillment from helping creatures under duress, these encounters seldom elicit positive affect. It is difficult to divorce the inherent need of any XR experience to foster positive user experience (UX) with the negative affect associated with exposure to the suffering of another being, human or otherwise. For example, research shows that individuals who regularly care for sick animals regularly report high levels of negative affect during and after interactions [56]. Considering the importance of the context in which the interaction is situated, this factor requires as much attention as the presentation of the species itself.

Should we lean towards facilitating simple, non-affective experience such as feedings or petting? This is not to say that simple encounters cannot leave an impression on users and build human-wildlife bonds. Recall the scene in Jurassic Park where the group encounters a sick Triceratops? Merely touching and hearing the beast breathe struck awe into the scientists, stirring positive emotions connecting them to the species. Unfortunately, in this example, awe was undoubtedly triggered by the "small self", or the size difference between the human and the non-threatening target [52], along with the obvious novelty of it being an extinct species. I doubt that an American encountering a sickly East African Lowland Honeybee FfF in VR would be awe-struck. Nonetheless, we must account for

how these factors leave an impression on users which permeates through the rest of their lives, the implications of which we are only now beginning to understand.

4.2.4 First Contact: Expectations and Associations. For many users, XR-based interactions with wildlife represent their first "direct" engagement with a non-domesticated animal. As such, the nature of these experiences may lead to divergent associations between conservation actions and impact; what users do in the virtual world to benefit wildlife implicitly implies a standard that many users cannot meet through real-world actions – we all cannot go to Antarctica to wash oil-slicked penguins. It is understood that humans rely on previous experiences to shape expectations of both themselves and others [31]. Similarly, if a user's first conservation-related activity involves direct handling of wildlife in XR, these experiences shape future expectations about what users may feel is demanded of them as biodiversity stewards. If that expectation is set too high by a simulated interaction there may be a decline in actual conservation behaviors, including advocacy, donations, and volunteerism. Thus, first encounters must not be taken lightly.

4.3 Cleaning Penguins in AR

Where VR brings us to the frozen tundra of the Adelie penguin, AR allows us to bring the penguins into our living rooms. There are several reasons why this modality might appeal to conservationists. For one, recreating a virtual environment that accurately reflects a creature's habitat may be too expensive to complete in terms of cost and time. Second, depending on the platform, (mobile-based) AR platforms are easier to use and more widely accessible compared to VR platforms, though less immersive. With this in mind, we sought to adapt the VR penguin simulation to an AR, albeit with a focus on HMD-based AR. Built using Unity 3D software for the Magic Leap HMD, the experience allows users to see a virtual, animated oil-slicked Adelie penguin in their environment. Using hand-tracking, we were able to detect specific gestures, allowing the user to wave and call over the penguin. After the penguin waddles to the user's location, the user is then able to pick up the penguin using both hands, set it atop a surface, and begin cleaning it. Once the penguin is clean (Figure 1), users can place it back on the floor so that it may exit the room and return to its "natural habitat". While preliminary

data suggests that these experiences can have a similar impact on users as it relates to human connectedness with the species, AR-based interactions are no less subject to critique as their VR counterparts.

4.3.1 People, Places, and Penguins. The convenience of being able to interact with wildlife in the comfort of your own home seems sensible on the surface, but home means different things to different people, and there is much to account for when bringing virtual beings into our personal spaces. Research on place attachment asserts how we form emotional connections to physical spaces (e.g., our home), and these connections in turn shape our affective and cognitive responses to interactions occurring in those spaces [64]. As emotion-eliciting stimuli, our personal spaces can spillover and influence how we feel and evaluate things within those spaces as well, a phenomenon known as emotional contagion [26]. Thus, if our interactions with wildlife become situated in our personal spaces, designers much account for how individual attachments to those spaces may shape connection to the exemplar and the species at large. We may come to love the penguin when we see it in grandma's house, and dislike the penguin because we encountered it in a classroom we're not fond of. Moreover, situating experiences in developed environments (e.g., apartments, labs) takes the "wild" out of wildlife encounters. This may prove problematic considering that the perceived exoticism of places stir interest and engagement [50] which may otherwise be tempered when wildlife interactions occur in familiar spaces. In sum, the interplay between people, places, and penguins cannot be overlooked.

4.3.2 Programming Animal Behavior. How do we decide how wildlife should respond and treat us in XR? Generally, wild animals respond to the mere presence of humans with fear and avoidance behaviors [23]. Scholars note the problematic nature of habituating wildlife to humans [24], but what of habituating humans to wildlife? If users are exposed to pleasant, albeit unrealistic, interactions with wildlife, and these encounters contribute to the individual's mood and well-being, motivations for real-world interactions may emerge. Mediated experiences with wildlife may establish unrealistic expectations, leaving many disappointed with the reality that most wildlife want nothing to do with us. In fact, such individuals may be disappointed to find that FtF encounters with wild animals (e.g., feedings) may result in aggressive behavior and attacks [14], a far cry from the docile, pleasant encounter that will most likely be presented in XR. Thus, much like media representations of wild animal behavior shape audience expectations of their real-world disposition (e.g., docility) [61], we can expect this effect to be more pronounced in XR, where interactions with wildlife are perceived as lived experiences rather than moving images on a screen. Design of XR-based wildlife interactions shape the narrative around human-wildlife relations, and thus require more thoughtful consideration as it relates to potential implications in the real world.

4.4 Snapchat and Loggerhead Sea Turtles

In partnership with Snapchat, we sought to adapt our wildlife experiences to more scalable, accessible AR platforms. Snapchat currently has over 200 million daily active users [76], and has been a driving force of AR adoption for years. As such, we created Project

SHELL (simulating habitat experiences of living loggerheads) as an initiative to connect audiences to the plight of Loggerhead sea turtles (*carreta carreta*) with the goals of encouraging conservation action. Project SHELL began as a VR experience, which we elaborate upon in the next section, and was subsequently adapted to fit within the Snapchat ecosystem. The AR experience, or Snap Lens, was created using Lens Studio and allows users to plant a sea turtle nest in their homes (Figure 2). After planting their eggs, a sea turtle hatchling emerges. The user must then help the hatchling navigate to shore. Narration contextualizes the experience by describing the various threats facing sea turtles at this stage in their lives. Upon saving the turtle, users are then able to snap a selfie with the hatchling they just saved and share their experience with friends. While the lens was well-received and used by hundreds of users to date, there are potential negative implications associated with this form of interaction that warrant exploration.

4.4.1 Representation Matters. On its own, the scale and ubiquity of Snapchat make it an optimal conduit for global audiences to access threatened wildlife in their own spaces. However, we must also take into account how facilitating these otherwise rare encounters may influence how users perceive and care about their local wildlife. Indeed, Battisti argues that communication of exotic and charismatic wildlife may contribute to conservation efforts at the expense of local species, and that while engaging with Pandas may be enjoyable and important, connection with local biodiversity cannot be overlooked [9]. That is, we may expend our empathy and advocacy to those species with whom we have connected with via XR as opposed to those in our own backyard that may otherwise be small or out-of-reach.

Speciesism undoubtedly exists and can be exacerbated through XR-based interactions. We must ensure that XR content does not limit its content to flagship species: animals that are attractive, large, entertaining, or useful [67]. Local wildlife may not be as exciting or exotic, though these species undoubtedly require our support, and by extension representation in our virtual spaces. Can we leverage principles of personalization to create human-wildlife interactions tailored to the user's contextual environment? Indeed, we know that experiences like *Pokemon Go!* presents creatures differentially based geographic factors unique to the user's location. If personalization yields greater affinity for the content, in this case wildlife, perhaps XR-based wildlife experiences should prioritize *local nature*, rather than just flagship species. The fact that local species are less known compared to exotic ones despite their proximity to audiences [6] suggests that such efforts would be beneficial.

4.4.2 Selfies Over Suffering. It is natural for humans to share information that they are proud of. Research on volunteer tourism show that individuals have a tendency to actively seek out images with the beneficiaries of their work; selfies with underserved and otherwise vulnerable communities is a common feature of Western goodwill [65]. As Schwarz and Richey articulate in their critique of social media's function in volunteer tourism, this practice is problematic because it (a) provides individuals with a false sense of contribution (e.g., slacktivism), and (b) degrades vulnerable communities and propagates negative stereotypes of groups depicted alongside the volunteer. In much the same way, AR-based interactions that involve the user aiding threatened wildlife may lead to

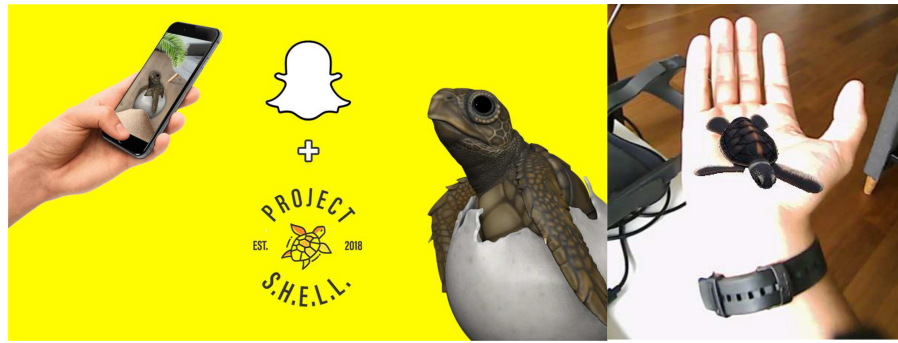


Figure 2: Project SHELL partnered with Snapchat (left) to create an AR lens about sea turtle conservation (right).

similar problematic outcomes, especially when they occur within AR platforms that are inherently social, such as Snapchat. Wildlife organizations note how social media content containing wildlife can spur others to seek selfies with wildlife, which can have detrimental effects on biodiversity conservation efforts and the survivability of a species in question [25, 39]. As noted in [39], individuals who viewed viral videos of a person tickling an endangered slow loris (*Nycticebus spp.*) tended to express a desire to obtain the species, “without demonstrating awareness of the risks to slow loris welfare”. While AR may provide a proxy for in vivo interactions with endangered wildlife, spreading videos of users in direct contact with virtual species that otherwise would not benefit from such an interaction may elicit similar responses that undermine the original goal of biodiversity conservation.

Scholars also warn of a “shifting baseline” issue related to when technological nature replaces real nature, even in cases where it is warranted (e.g., extinct species) [28]. Shifting baseline refers to how humans may judge issues as less severe as they become normalized to those states. If future generations are constructing their conception of wildlife through technology, there will certainly be bias, bias which may lead to inaccurate perceptions of the species, the plights they face, and their severity, all of which shapes their relationship with the natural world. Put simply, if threatened (virtual) wildlife are increasingly seen through platforms like Snapchat, it is likely that audiences may erroneously believe the species count to be numerous. Indeed, preliminary research conducted through Project SHELL indicates that exposure to many versus a single endangered marine mammal leads people to underestimate the species’ extinction classification.

4.4.3 Seeking Out Wildlife. Please, if you’re reading this, do not seek out penguins or sea turtles in the wild. Yes, an inherent goal with most of the aforementioned experiences is to build connection with wildlife. As with any bond formed between beings, the attraction often manifests itself in a desire for direct interactions. On the surface this may seem innocent enough, though a potential concern relates to how AR-based encounters may set a precedent for how humans believe they can (or should) engage wildlife IRL. As [74] notes, “harassment of marine mammals occurs when individual members of the public attempt interactions on their own”, leading to instances where humans fail to respect the boundaries of species, creating undue stress, and ultimately leading to greatly

reduced welfare and death. In the case of sea turtles, instances of “selfie mobs” attempting to interact with nesting turtles led to the abandonment of various nests [59]. As good-intentioned as Project SHELL may be, can AR-based interactions with sea turtles lead users to seek out direct experiences IRL to the detriment of the species? These are all very real possibilities which must be further investigated.

5 INTERACTIONS AS WILDLIFE

If this is possible for a badger, perhaps it is possible for us. Perhaps even for me. Perhaps if we all shuttle enough across frontiers the world won’t fall apart – p. 71.

In Charles Foster’s book “Being a Beast”, he documents his time living amongst wildlife, including badgers, otters, and foxes. These embodied experiences, he argues, help bridge the human-wildlife gap through CWN. Yet, one need not burrow into soil, munch on grass, or brave the elements alongside beasts to embody their lived experiences. Modern VR set-ups are notably capable of creating the illusion of BT with non-human bodies with anatomical structures different from our own [3, 35, 36, 68]. This is an incredibly compelling, and convenient, alternative to Foster’s approach, and one which served as the basis for Project SHELL’s initial VR component.

Project SHELL originally began as an exploration into VR-based animal embodiment, and how it can be used to advance biodiversity conservation efforts by connecting users to the lived experiences of threatened species. The simulation, which is now being ported to the Oculus Quest, begins by allowing users to enter a sea turtle nest, selecting an egg, entering the egg, and being reborn as a hatchling. After breaking out of their shell, users are then tasked with navigating to the open water, avoiding various threats along the way (e.g., light pollution, traffic). Afterwards, users become adult sea turtles and must find food in the open water, avoiding boats, marine debris, and netting. Lastly, users progress to nesting adults where they must return to shore and complete the cycle by finding a suitable nesting site, during which they must avoid threats such as shore armoring.

As shown in Figure 3, Project SHELL enables visuomotor synchrony by reflecting the user’s head and limb movements accurately onto the sea turtle avatar. Moreover, using a custom-designed seating apparatus, users are able to assume the posture of a non-bipedal



Figure 3: Project SHELL users in controlled lab settings (center) and field study (right) embodying an injured Loggerhead sea turtle (left).

creature. A haptic backpack functions as the user’s shell, vibrating based on various collisions during gameplay. Several studies have been conducted using Project SHELL. Overall, animal embodiment is a promising alternative to contemporary forms of environmental storytelling. Results demonstrate that the magnitude of BT with a turtle body can influence pro-environmental attitudes and actual monetary donations towards sea turtle conservation. However, pressing ethical questions arose over the course of several controlled lab and field studies, some of which are outlined below.

5.1 Becoming a Sea Turtle

5.1.1 Embodied Trauma. Project SHELL users resurfacing for air while exploring the virtual waters do so cautiously, as boats and netting serve as looming threats to their in-game survival. By design, users must encounter one of these threats, and when they do there is a visceral reaction due to the combination of audiovisual and haptic stimulation that accompanies the collision. This undoubtedly increases threat perceptions and connects them with the threat in a way a 2D alternative cannot. While perceiving threats as more severe, imminent, and self-relevant can contribute towards pro-environmental outcomes [15], there is the real possibility that these experiences can cause trauma. This is especially true of younger users. Because age is inversely related to one’s sensitivity to the BT illusion [16], and brain development is especially susceptible to environmental stressors during childhood [38], having children embody threatened wildlife may be ethically dubious depending on the nature and presentation of the threats.

5.1.2 Embodiment Does Not Guarantee Empathy. VR is considered an empathy machine in part because it can facilitate BT with distant others. Indeed, through illusory body ownership users literally put themselves in another’s shoes. Yet, emerging work suggests that the use of virtual embodiment to connect disparate groups can backfire depending on the context of the interaction. For example, when users embody another human of a different race in a negative setting, this experience can actually increase implicit racial bias [7]. Moreover, attitudes towards the embodied group can sour if individuals are merely exposed to the issues affecting the group, as opposed to pairing that exposure with additional information about the lived experience of the other [1]. Similarly, the potential stress involved with being thrust into an unfamiliar environment to cope with species-specific threats may backfire, leading users to seek distance rather than connectedness with the non-human

victim. Thus, we must approach the use of animal embodiment and its relationship to conservation outcomes cautiously. Embodiment on its own cannot guarantee connectedness and concern for the species.

6 WHERE DO WE GO FROM HERE?

Professional biologists don’t like talking about animal emotion. Chat about an animal as a piece of hardware running (or even being) a bit of software, and you’ll meet only smiles [21] – p. 39-40.

“Illusion that is coherent is called truth” [51], and the illusory equivalence between in vivo and XR-based interactions with wildlife can influence the human-nature gap, for better or for worse. This paper outlines how experiences with wildlife in XR may indeed advance biodiversity conservation, while highlighting valid concerns related to ethics, human health, and ecological well-being. My goal with this critique is not to detract from potential of XR in advancing environmental initiatives, rather it serves as a cautionary tale, one that we as HCI scholars may revisit when creating and assessing XR-based experiences with wildlife. One obvious starting point is to ask, “who are we designing for?” Indeed, CHI has a rich history of human-centered design work that explores this question across various verticals [49]. However, if I am to advocate for anything it is for increased consideration of wildlife-centered design. That is, XR content featuring non-human life must go beyond ensuring optimal UX for the human user, and instead expand the scope of consideration to include the residual impact the experience may have on the species in question. Simply including these considerations may result in alterations to virtual wildlife, such as changes to the animations, interactions, appearances, story, and context, among others. Hopefully, engaging in thoughtful design of XR-based wildlife experiences can help ensure that these encounters, synthetic they may be, benefit the user as much as they do the depicted species.

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